### 6.2.2 Stochastic Integration

In order to develop good sampling paths for the Gibbs-Boltzmann distribution, we need to introduce a method of adding random perturbations into a differential equation. Let us first consider the definition of the integral *with respect to Wiener measure*.

Recall that the Riemann integral of a continuous function $g$ on $[0, \tau]$ is the limit of Riemann sums constructed as follows: partition the interval $[0, \tau]$ into $K$ subintervals each of width $\delta t = \tau/\nu$, choose an arbitrary $t_k$ from the $k$th subinterval, $1 \leq k \leq \nu$, and evaluate the integral as a limit of the sum of products:

$$\int_0^\tau g(t)\mathrm{d}t = \lim_{\nu\to\infty} \sum_{k=0}^{\nu-1} g(t_k)\delta t.$$

We know that this limit does not depend on where the $t_k$ are chosen within the subintervals. For example we can take the left or right endpoint of each subinterval or the midpoint.

By analogy, we would like to write the stochastic integral of a function $g$, where $g$ may, itself, be defined in terms of a stochastic process, as the limit of a sum:

$$\sum_{k=0}^{\nu-1} g(t_k)[\mathrm{W}((k+1)\delta t) - \mathrm{W}(k\delta t)], \qquad \nu \to \infty,$$

where the increments $\mathrm{W}((k+1)\delta t) - \mathrm{W}(k\delta t)$ identified here are the increments of a Wiener process whose properties are given above. (In particular they are normally distributed with mean zero and variance $\delta t$.) First, we should be careful regarding the sense in which we expect convergence, since sequences of random variables may converge in various ways.

Next, unlike in the case of the Riemann sums, it turns out that the result depends on where we place the points $t_k$. Using the properties of the Wiener process we may easily show that the choice $t_k = k\delta t$ (left endpoint) and the choice $t_k = (k + 1/2)\delta t$ yield different formulas for the integral. The first choice is commonly referred to as *Itō integration*, and the second as *Stratonovich integration*. As an example, consider the case where $g(t) = \mathrm{W}(t)$. We suppose this to be approximated by a sum

$$\begin{aligned}\int_0^\tau \mathrm{W}(t)\mathrm{dW}(t) &\approx \sum_{k=0}^{\nu-1} \mathrm{W}(k\delta t)[\mathrm{W}((k+1)\delta t) - \mathrm{W}(k\delta t)], \\ &= \frac{1}{2}\sum_{k=0}^{\nu-1}\Big[\big(\mathrm{W}((k+1)\delta t)^2 - \mathrm{W}(k\delta t)^2\big) - \big(\mathrm{W}((k+1)\delta t) - \mathrm{W}(k\delta t)\big)^2\Big].\end{aligned}$$